The Circuit Survives, the Choices Collapse: Induction Heads under Recursive Synthetic Training

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Abstract

We train two-layer Transformers on the in-context symbol–value task of Singh et al. and recursively retrain each generation on labels sampled from its predecessor. A pilot run with a short training budget never left the pre-induction loss plateau, so we first establish that induction heads form at generation 0 (100% accuracy, induction score ≈1, causal head-ablation drops up to 70 points). In the base task (predict one value) we observe no collapse over five generations. In the extended task (also generate a next symbol and its value), consistency falls linearly from 99.4% to 94.8%, the training-label distribution skews towards particular symbols, and probability mass on symbols absent from the context grows from 0.7% to 4.9%. Retrieval, however, stays intact: whenever the generated symbol is in context its value is correct ≥99.6% of the time, and induction attention remains at ceiling. Collapse here acts on the model's free choices, not on its in-context learning circuit.

1 Introduction

In-context learning (ICL) lets a Transformer infer a mapping from examples in its prompt without weight updates [2]. A well-understood mechanism for it is the induction head [2, 5]. Model collapse is the progressive degradation of models trained on data generated by earlier models [3, 4]. As synthetic text accumulates online, it matters whether collapse damages the *mechanisms* of ICL or only the distribution of what models produce. We ask: *under recursive training, does the induction circuit degrade, and if not, where does collapse act?* Using a task where the circuit is known, we (i) show that a short training budget yields a non-induction heuristic and must be ruled out before studying collapse; (ii) find no collapse when the target is fully determined by the context; and (iii) localise collapse in the extended task to a growing context-independent symbol prior that bypasses an intact induction circuit.

2 Background

Induction heads implement "find the previous occurrence of the current token and copy what followed it". In a two-layer model a *previous-token head* in layer 1 writes each token's predecessor into its residual stream, and an *induction head* in layer 2 uses the current token as a query to match those keys and copies the following token's value [2]. Singh et al. [5] showed on this symbol–value task that the circuit forms through an abrupt phase transition after a long loss plateau, with several heads interacting. Shumailov et al. [3, 4] showed that recursive training compounds sampling and approximation errors, so learned distributions lose their tails and drift from the data. Circuit-level evidence should be causal rather than read from attention maps alone [1].

3 Method

Task. Each sequence draws a fresh random bijection between 8 symbol tokens and 8 value tokens. The context is `BOS`, six symbol–value pairs (symbols sampled with replacement), and a query symbol that appeared earlier, so the answer is only available in context. The *base* target is the query's value. The *extended* target is that value, then a next symbol, then that symbol's value. The next symbol is drawn uniformly from the distinct symbols shown. Our pilot drew it from all 8 symbols, which made its value unknowable in 45% of examples. The redesign keeps every target inferable while the symbol choice has a known uniform distribution against which collapse can be measured.

Model and training. A causal Transformer with 2 layers, 4 heads, width 64, MLP width 128, pre-LayerNorm, learned positions and tied embeddings, trained with AdamW (weight decay 10^{-4}, batch 128) on token cross-entropy over the target positions. We use 50k/2k/2k train/validation/test sequences, keep the checkpoint with the lowest validation loss (max. 20 epochs, patience 6), and select hyper-parameters on validation data only over learning rate $\{3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}\}$ and training size $\{3\text{k}, 50\text{k}\}$.

Recursion. Generation 0 trains on real labels. Generation $g{\geq}1$ is a freshly initialised model trained on the same contexts with labels sampled at temperature 1 from generation $g{-}1$ (synthetic fraction 1). Validation and test labels are always real, so evaluation cannot collapse with training. We run 6 generations × 2 tasks × 3 seeds (36 models).

Metrics. Behaviour is measured on clean test data both teacher-forced and free-running. For extended outputs sampled at $T{=}1$, *pair consistency* is the fraction whose generated symbol occurs in the context *and* whose next value matches it. Exact match is uninformative here: the next symbol is random by design, capping it near 0.24. Distribution drift is the most common symbol's share of training labels (uniform is 12.5%). Mechanistically, the *previous-token score* of a head is its mean attention from position t to $t{-}1$, and the *induction score* is the attention mass from the query to the value tokens following earlier occurrences of the query symbol. A head attending uniformly over value tokens scores 0.27. For causal evidence we zero each head's output in turn and record the drop in first-value accuracy. Finally, *leaked mass* is the probability the model assigns, at the symbol step, to symbols not shown in the context.

4 Results

Pilot: a plateau, not induction. With 3k examples (≈600 steps) generation-0 accuracy was $32.0 \pm 3.1\%$, between the heuristics "copy a random context value" (25.8%) and "copy the most frequent value" (36.8%). Its best induction score (0.24)

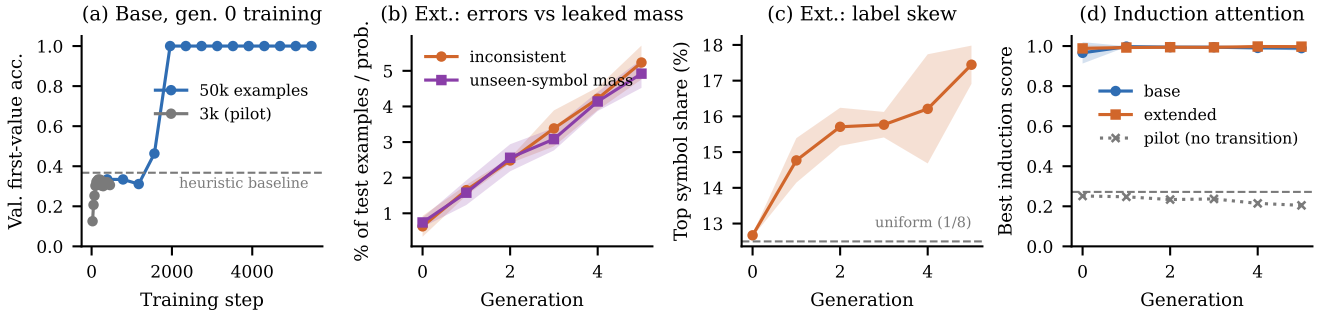


Figure 1: (a) Generation-0 validation accuracy: with 3k examples training stops on the plateau near a copy heuristic, while with 50k the induction transition occurs at ≈1,600–2,000 steps. (b) Extended task: inconsistent sampled pairs and probability mass on unseen symbols grow together, ≈0.8–0.9 points per generation. (c) The most common symbol's share of training labels rises above uniform. (d) Induction attention stays at ceiling in both tasks; the pilot never left the uniform-attention baseline (dashed). Bands: ±1 SD over 3 seeds.

Table 1: Clean-test results at generations 0 and 5 (mean ± SD over 3 seeds).

Task	Metric	Gen. 0	Gen. 5
Base	First-value accuracy	1.000 ± 0.000	1.000 ± 0.000
Base	Best induction score	0.966 ± 0.054	0.990 ± 0.014
Extended	First-value accuracy	1.000 ± 0.000	1.000 ± 0.000
Extended	Sampled pair consistency	0.994 ± 0.003	0.948 ± 0.005
Extended	Leaked mass on unseen symbols	0.007 ± 0.001	0.049 ± 0.004
Extended	Top symbol share of training labels	0.127 ± 0.000	0.175 ± 0.005
Extended	Best induction score	0.988 ± 0.015	0.997 ± 0.002

was at the uniform baseline (0.27), so its apparent "collapse" was the decay of a heuristic. In the sweep every learning rate stayed at 30–34% with 3k examples. With 50k examples, 10^{-3} and $3{\cdot}10^{-4}$ reached 100% and $3{\cdot}10^{-3}$ stalled at 70%. We selected 10^{-3}, which crossed 99% fastest (1,955 steps), and Fig. 1a shows its plateau and abrupt transition.

Base task: no collapse. All 18 base models reach 100% test accuracy at every generation. Synthetic labels are ≥99.998% correct, so each generation relearns the same function. The best induction score stays at 0.90–1.00 and the best previous-token score at 0.53–0.63 (Table 1).

Extended task: collapse of the choice, not the retrieval. First-value accuracy is 100% throughout, yet sampled pair consistency declines linearly (-0.9 points per generation; Fig. 1b). Almost every failure is a symbol not in the context: conditional on the generated symbol being shown, its value is correct in ≥99.6% of cases at every generation, and greedy decoding stays 100% consistent. The failures come from probability mass leaking to unseen symbols (0.7%→4.9%), which closely tracks the error rate. The leak is not diffuse: one symbol receives 20–51% of leaked mass in each model, against 12.5% if spread evenly. The labels each generation trains on skew accordingly: the top symbol's share rises from 12.7% to 17.5% and entropy falls from 3.000 to 2.974 bits (Fig. 1c). The generated symbol's mean context position (0.41) matches real data, so no positional bias is amplified.

Mechanism stays intact. Induction scores stay at 0.97–1.00 and previous-token scores at 0.55–0.64 in every extended model (Fig. 1d). Ablating the single most important head lowers first-value accuracy by 9–70 points across all models, confirming a causal circuit. That head varies across seeds and generations (layer 1 in 24 of 36 models), consistent with redundant, re-learned implementations in freshly initialised models.

5 Discussion and limitations

Collapse appeared only where the target leaves the model a free choice. A deterministic in-context lookup is a fixed point of recursion: a perfect model emits perfect labels. A uniform choice among context symbols is not. Each generation places a little probability on a context-independent symbol prior, sampling turns that into training labels, and the next model learns those labels as legitimate data. Errors therefore accumulate roughly linearly instead of being corrected. Unlike the loss of tails in [4], the damage here is *spurious* mass that grows in the tail while the mode, and greedy and top-1 metrics, stay perfect. Since induction attention and causal ablations stay at ceiling, the circuit is not what degrades. The leaked mass must enter through a pathway that bypasses copying, such as direct token biases in the embedding–unembedding path; we have not yet traced it. Collapse can thus hide behind intact circuits and perfect accuracy; distributional metrics on sampled outputs are needed.

Limitations. Only labels are recursive (contexts stay real), with synthetic fraction 1 and $T{=}1$. Five generations, three seeds, an 8-symbol vocabulary and a 2-layer model limit generality. Early stopping on real validation data gives every generation a little real signal and may slow collapse. Attention scores are measured only at the query position, and single-head ablation understates redundancy. Future work: trace the leaked pathway with direct logit attribution or path patching, vary the synthetic fraction and temperature, and run more generations.

References

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Contribution statement

This is an individual submission. Sibonelo Mlambo (536187) carried out the experimental design, implementation, experiments, analysis, figures and writing.

Code and reproducibility

Code, configurations, raw results and checkpoints accompany the submission. `tune.py` reproduces the hyper-parameter selection, `run_study.py` the main study, `analyse_leakage.py` the leaked-mass analysis and `make_figures.py` Figure 1.

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Declaration of AI use (draft). Generative AI (Claude, Anthropic) assisted with code development and debugging, running and analysing experiments, producing figures, and drafting and editing this report. All experiments were run on the submitted code, and every reported number comes from the saved result files. I have reviewed the code and text, can explain them, and take full responsibility for the correctness and integrity of this submission.